

Gunpowder weapons

By the 13th century, knowledge of how to make gunpowder had spread beyond China to the rest of Asia and into Europe. People soon saw how this deadly invention could be used as a weapon. As armed forces began to use gunpowder in increasingly powerful and accurate firearms, warfare changed rapidly.

Earliest known picture of a fireball from a 10th-century Chinese wall painting



First fireball

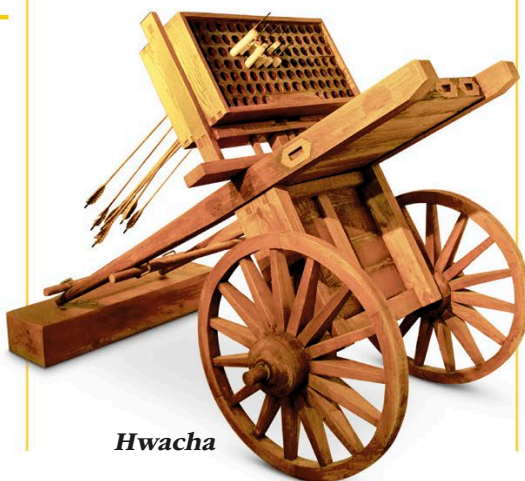
- **What?** Grenade
- **Who?** Medieval Chinese
- **Where and when?** China, 11th century

A fireball, or grenade, is a small bomb that can be thrown by hand. Early fireballs, first developed in China, were hollow clay or metal vessels filled with gunpowder. Attached to the fireball was a paper fuse, which was lit and left to burn down until it ignited the gunpowder, causing the weapon to explode.

Fire chariot

- **What?** Rocket launcher
- **Who?** Chinese
- **Where and when?** China, 14th century

Although the Chinese were the first to invent simple rocket launchers, the Koreans developed a more powerful weapon called the *hwacha*, or fire chariot—a two-wheeled cart on which was mounted a rectangular wooden frame. This frame was loaded with *singijeon*, or fire arrows, each carrying a gunpowder charge timed to explode when an arrow struck its target. The largest *hwacha* could launch 200 fire arrows all at once, at a range of 330–1,475 ft (100–450 m).

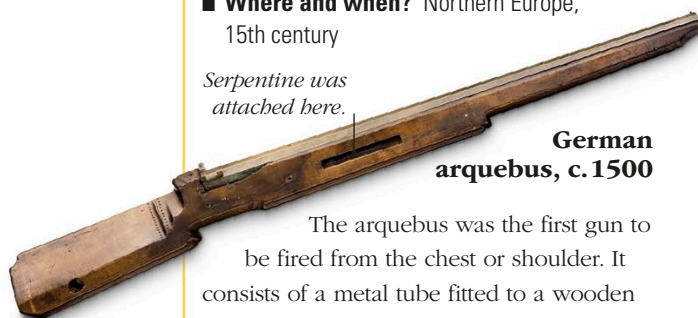


Hwacha

Arquebus

- **What?** Hook gun
- **Who?** Unknown
- **Where and when?** Northern Europe, 15th century

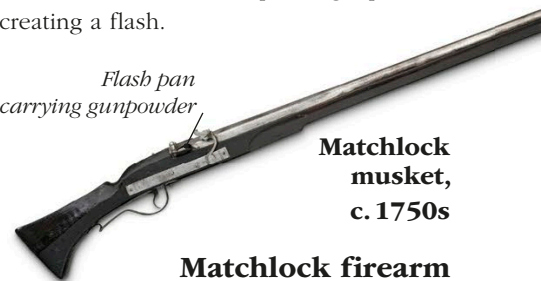
Serpentine was attached here.



German arquebus, c.1500

The arquebus was the first gun to be fired from the chest or shoulder. It consists of a metal tube fitted to a wooden stock, or handle. Its name comes from an Old French word meaning “hook gun,” because many models had a hook on the barrel to hold the weapon steady against a support. The gun was fired with an S-shaped lever called a “serpentine” that lowered a lit match into a pan of gunpowder, creating a flash.

Flash pan carrying gunpowder.



Matchlock musket, c.1750s

Matchlock firearm

- **What?** Arquebus with matchlock mechanism
- **Who?** Unknown
- **Where and when?** Northern Europe, c. 1475

The matchlock made firing weapons quicker. It was a trigger-controlled device that used a spring to move a lit match toward a gunpowder charge placed at the rear of the weapon. This set off a small explosion, which produced high-pressure gas that shot down the barrel, forcing out the ammunition.

WOW!

Reloading early firearms such as the musket took a long time: even the best-trained soldiers could fire no more than five times a minute.